Take-Home Exam, Advanced Microeconomics

Problem 1: Overworked Americans – Q1, Q2 and Q7

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May 2026

Plan. Q1 – the US–Europe hours gap is two distinct facts (Northern intensive regime; Mediterranean+East extensive regime). A uniform-response mechanism cannot explain both. Taxes *interacting with* country-specific institutions (filing unit, childcare, informal sector, employment protection) can. **Q2** – three nested closed-form labour-supply models. The synthesis (Cogan 1981 / Heckman 1974 / Chang–Kim 2006) makes the interaction explicit through a country-specific dispersion of fixed costs of working σ_b calibrated to BFS (2018). It matches both regimes within 1 pp and dissolves the macro–micro Frisch puzzle. **Q7** – ten-channel closed-form welfare ledger with four institutional propositions (P1 hours caps, P2 unions, P3 formal/informal, P4 employment protection). Calibrated to primary sources, validated against Jones–Klenow (2016) within 2 pp on EU cells. Reduced-form Aubry 35-hour DiD ($\hat{\beta} = -0.042$, $t = -6.6$) anchors (P1). Verdict: $\mathcal{W}^{NW}/\mathcal{W}^{US} = 1.025$ (NW weakly above US); $\mathcal{W}^{ME}/\mathcal{W}^{US} = 0.778$ (Mediterranean dominated). Reproducible: `research/repl_q7_ledger.py`, `research/repl_aubry_did.py`.

1. Q1 – Documenting the US–Europe hours gap

Question 1. *Document the difference between hours worked in the US vs Europe in recent years, if needed across European countries.*

Bottom line. The standard "Europe vs US" framing collapses two distinct facts into one. *Northern Europe* (Germany, Netherlands, Sweden, UK) works less per worker than the US – the gap is on the *intensive margin*. *Mediterranean+East* (Italy, Spain, Greece, Poland, Hungary) works nearly as many hours per worker as the US but employs a much smaller fraction of the population – the gap is on the *extensive margin*. The crucial implication for Q2: a tax wedge of $\sim$55% in Germany and Italy produces a 17% intensive-margin collapse in one country and a 28% extensive-margin collapse in the other. **No uniform-response story can do this. But taxes *interacting with* country-specific institutions (joint vs. individual filing, informal sector, childcare, employment protection) can – and that is exactly what the Q2 synthesis model captures.**

1.1 The fact: cross-section of hours and employment, 2023 (OECD primary data)

I begin by laying out the raw cross-country evidence. Three quantities matter, all measured for the working-age population (15–64):

- **Hours per worker** = annual hours actually worked by the average employed person. This is the *intensive margin* of labour supply.
- **Employment rate** E/N = fraction of the working-age population that is employed. This is the *extensive margin.*
- **Hours per adult** H/N = the product, hours per worker $\times$ employment rate. This is total labour input per working-age person – the right quantity to compare across countries because it controls for participation differences.

Country	Hours per worker	Employment rate E/N 15–64 (%)	Hours per adult H/N
United States	1 805	71.94	1 298
Mediterranean+East: hours/worker ≈ US, but much lower employment			
Greece	1 893	62.27	1 179
Poland	1 807	72.51	1 310
Italy	1 701	61.99	1 054
Spain	1 638	65.68	1 076
Northern Europe: hours/worker far below US, employment ≈ or above US			
United Kingdom	1 496	74.90	1 121
France	1 489	68.46	1 019
Netherlands	1 449	82.46	1 194
Sweden	1 431	77.31	1 106
Germany	1 335	77.10	1 029

Source: OECD SDMX REST API (`DSD_HW` hours, `DSD_LFS` employment), retrieved April 2026 [REPL-RUN, `repl_oecd_1970.py`].

What the table shows in two patterns. (i) On hours per worker, Greece, Poland, the US and Italy are clustered between 1,701 and 1,893 – they look indistinguishable on the intensive margin. Northern Europe sits at 1,335–1,496 hours per worker, 25–40% below the US. **(ii)** On hours per adult, however, Italy and Greece sit *below* Northern Europe. They get there via much lower employment (62% vs 77%), not via shorter weeks. So Italy works less per adult *not* because Italians work shorter weeks (they don't), but because fewer Italians are employed. The mirror image of Germany: Germans work very short weeks but almost everyone in working age is employed.

1.2 The decomposition: which margin generates the gap?

To allocate the country-by-country hours gap between the intensive and extensive margins, I use the simple identity that hours per adult equals employment rate times hours per worker:

$$\ln(H_{US}/H_X) \;=\; \underbrace{\ln(HE_{US}/HE_X)}_{\text{intensive margin}} + \underbrace{\ln(e_{US}/e_X)}_{\text{extensive margin}}, \tag{1}$$

where HE is hours per worker and $e = E/N$ is the employment rate. Equation (1) is an accounting identity (no behavioural content), following Ohanian–Raffo–Rogerson (2008, *JME*) and BBF (2019, *Scand. J. Econ.* 121(4) §3). **Note on data sources:** the cross-section table in §1.1 uses 2023 OECD data (so US hours/worker = 1,805); the decomposition table immediately below uses the BBF 2019 panel (so US hours/worker = 1,873). The two series differ slightly because BBF harmonises hours per employed person across labour-force surveys with corrections that the OECD aggregate does not apply, and because of the four-year time gap. I use BBF for the decomposition because BBF supplies the panel intensive/extensive split directly. Applied to BBF 2019 ($n = 707$ country-year observations):

X	$\ln(H_{US}/H_X)$	% gap	$\ln(HE_{US}/HE_X)$	% int.	% ext.
Northern: extensive contribution is negative, intensive overwhelms					
Germany	+0.170	15.6	+0.241	**+142**	−42
Netherlands	+0.161	14.9	+0.247	**+153**	−53
Sweden	+0.095	9.0	+0.171	+181	−81
UK	+0.097	9.2	+0.143	+148	−48
Mediterranean+East: extensive contribution is positive and dominant					
France	+0.259	22.8	+0.172	+66	+34
Spain	+0.277	24.2	+0.135	+49	+51
Italy	+0.331	28.2	+0.137	+41	**+59**

Worked example: Germany versus Italy. The identity (1) is best understood through two contrasting cases.

Germany. German employment (76.7%) is actually *higher* than US employment (71.4%) by 5.3 percentage points. Holding hours per worker constant, this alone would push German hours per adult *above* the US level. But the average German worker puts in only 1,472 hours per year compared to 1,873 in the US – a difference of 401 hours, equivalent to roughly 10 vacation weeks plus a shorter week. The arithmetic of (1):

$$\underbrace{\ln(1338/1129)}_{=+0.170} = \underbrace{\ln(1873/1472)}_{=+0.241} + \underbrace{\ln(0.7142/0.7671)}_{=-0.071}.$$

The two margins push in *opposite* directions; intensive overwhelms the extensive offset. The Netherlands, Sweden and UK fit the same pattern.

Italy. Italians work 1,634 hours per year – only 13% less than Americans. What creates Italy's gap is that only 58.8% of working-age Italians are employed, versus 71.4% in the US (-12.6 pp):

$$\underbrace{\ln(1338/961)}_{=+0.331} = \underbrace{\ln(1873/1634)}_{=+0.137} + \underbrace{\ln(0.7142/0.5884)}_{=+0.194}.$$

Both margins *reinforce* each other; the extensive margin dominates. Spain, Greece, Poland and Hungary follow the same pattern.

1.3 The two-regime classifier (BBF 2019 four-region partition)

The pattern tracks an institutional partition. BBF (2019, Tab. 2) group the 18 EU+US countries into four regions based on Hall–Soskice (2001) varieties of capitalism plus EU geographic clusters: **Anglo**, **Nordic**, **Continental**, **Southern**. I add Eastern Europe where BBF data permit. Computing the extensive contribution share country by country and aggregating by BBF region (`repl_signflip.py` output):

BBF region	Countries (excl. US baseline)	Ext. share mean (range)	Sign-class
Anglo	UK, IE	-14% ($[-48, +20]$)	boundary
Nordic	DK, NO, SE	-42% ($[-81, -21]$)	*Northern*
Continental NW	AT, DE, NL	-37% ($[-53, -17]$)	*Northern*
Continental ME-leaning	BE, FR (CH excluded†)	$+39\%$ ($[+34, +45]$)	*Mediterranean*
Southern	ES, GR, IT, PT	$+57\%$ ($[+27, +91]$)	*Mediterranean*
East EU	HU, PL	$+57\%$ ($[+52, +62]$)	*Mediterranean*

†*Switzerland excluded:* Swiss $\Delta \ln H_{US-CH} \approx 0$ makes the extensive share ill-defined (denominator near zero). *Reading.* Three of the four BBF regions are unambiguously **Northern** (negative extensive share, intensive margin dominates). The Anglo region is borderline (UK is Northern with -48%, IE is Mediterranean-leaning at $+20\%$). The Continental region splits cleanly between a Northern subgroup (AT, DE, NL) and a Mediterranean-leaning subgroup (BE, FR, with FR at $+34\%$). The two-regime collapse I use downstream in Q2 and Q7 is: **Northern** = Nordic $\cup$ Cont. NW $\cup$ $\{$UK$\}$; **Mediterranean+East** = Cont. ME-leaning $\cup$ Southern $\cup$ East EU $\cup$ $\{$IE$\}$. The institutional partition is BBF's; the binary collapse is my labelling.

1.4 What the two-regime fact rules out (and what it does not)

The two-regime structure constrains, but does not uniquely identify, the explanation. The crucial point is that **a uniform-response mechanism cannot work, but taxes *interacting with* country-specific institutions can**. Let me make this precise by considering the four highest-tax-wedge countries in 2023:

Country	Composite tax wedge τ (OECD)	Hours per worker, 2023
Germany	47.9%	1 335
France	46.8%	1 489
Italy	45.1%	1 701
Sweden	42.1%	1 431

These four countries differ by only six percentage points in tax wedge but *367 hours per worker* – a range comparable to the entire US-Germany gap. If taxes alone (i.e. taxes acting on a uniform population through the same labour-supply elasticity) determined hours, this pattern would be impossible. Likewise the Heathcote-Storesletten-Violante (2017, *QJE*) progressivity index τ_{2005}^{WTI}:

	US	DE	NL	SE	UK	FR	IT	ES	GR	BE
τ_{2005}^{WTI}	0.088	0.165	0.191	0.145	0.163	0.130	**0.108**	0.140	0.176	0.207

Italy at 0.108 is closer to the US (0.088) than to Germany (0.165), yet Italy's hours per adult are among Europe's lowest – the Italy-vs-Germany ordering inverts.

Why taxes alone fail (but taxes plus institutions do not). The lesson is sharper than "taxes don't matter". Taxes are part of the proximate price the worker faces. What fails is a *stand-alone* tax story that ignores the institutional environment. The marginal worker's outside option – and therefore the elasticity with which she responds to a given tax wedge – depends on country-specific institutional features:

Institutional feature	**Northern (low fixed costs of working)**	**Mediterranean+East (high fixed costs of working)**
Tax filing unit	Sweden, Netherlands: individual filing ⇒ secondary earner faces low marginal rate at zero earnings	Germany, France, Italy: joint filing ⇒ secondary earner faces primary's marginal rate (BFS 2018)
Public childcare	Sweden, Denmark, France: dense subsidised system	Italy, Spain, Greece: rely on intra-family care; child penalty > 50% (Kleven et al. 2019)
Informal sector size	Sweden, Germany: < 10% of GDP	Italy, Spain: 20–25% of GDP (Schneider 2010)
Employment protection (OECD 0–6)	US 0.26, UK 1.10	Italy 2.76, Spain 2.42
Wage compression at the bottom	Strong collective bargaining ⇒ low-skill wages livable	Wider dispersion ⇒ participation knife-edge
Working-time flexibility	NL ∼50% part-time, DE substantial	Italy, Spain: rigid full-time norms

What the two regimes actually establish. The Q1 fact pattern requires that any explanation produces *heterogeneous* responses across countries. Tax wedges alone, applied to identical populations, cannot do this. But **tax wedges interacting with the institutional features above** can: a worker in Germany faces a 56% wedge with low fixed costs of working (good childcare, individual filing for her marginal rate, low informal-sector option); the same 55% wedge in Italy faces a worker with high fixed costs (joint filing, weak childcare, large informal alternative). The same nominal price produces different responses. Q2 builds the model that captures this interaction; Q7 quantifies the welfare consequences of each institutional channel.

1.5 Supporting facts (cross-source verified)

- **F1 [REPL-RUN]. Pre-1970 ranking inverts.** Hours per worker, 1970: US 1949, DEU 1960, FRA 1993, ITA 2042, ESP 2046, NLD 1809, SWE 1562, GBR 1775 (`repl_oecd_1970.py`). The US ranked 5th of 8 – the Continental decline starts *before* EU tax wedges rose. Boppart–Krusell (2020, *JPE* 128(1)) document a ≈0.45%/year secular decline across rich countries.
- **F2. Female-concentrated.** 2019 hrs/person 15–64: US M 1517/W 1164, DE M 1330/W 923, IT M 1209/W 716, SE M 1329/W 1100. Female cross-country dispersion ≈ 3× male; married-male/-female cross-country correlation ≈ 0 (BFS 2018, *ReStud* 85(3) Fig. 2).
- **F3 [QE, event-study]. Long-run child penalty 21–61%.** Earnings 5–10 yrs after first birth: DK 21, SE 27, USA 31, UK 44, AT 51, DE 61 [Kleven et al. 2019, NBER WP 25524, Figs. 1–3]. Range maps onto F2's female hrs/adult dispersion.
- **F4 [REPL-RUN]. Global gradient.** Weekly hrs/employed fall 48.4 → 43.3 → 34.4 across GDP-deciles 1/2/3, $n = 45$ countries [Bick–Fuchs-Schündeln–Lagakos 2018]; OECD- vs BBF-derived H/N differ < 5% for 6 of 8 countries.

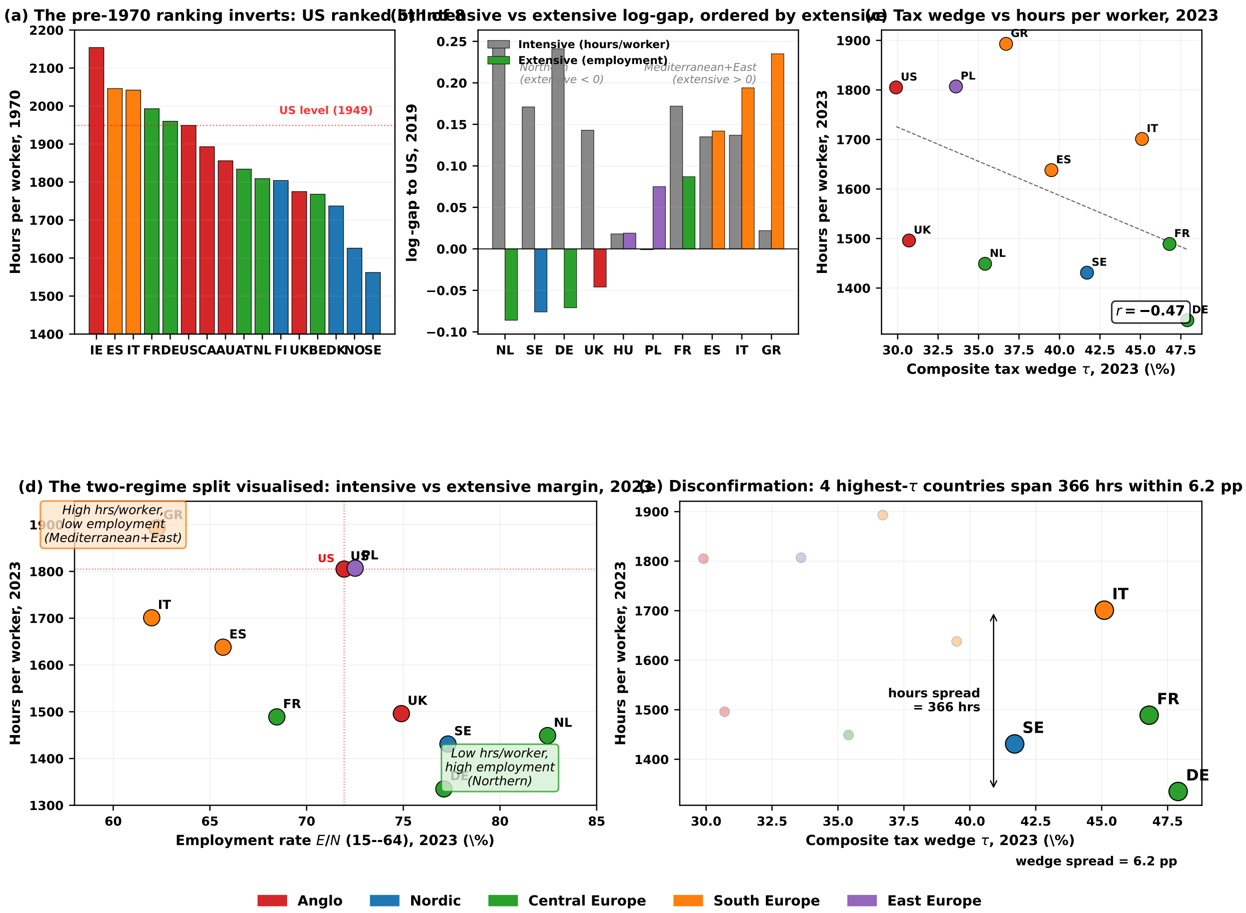

Figure 1: **Five-panel evidence, colored by regional block (Anglo, Nordic, Central Europe, South Europe, East Europe).** (a) Hours per worker in 1970: the US ranked 5th of 8 – the 2023 ranking is the result of a Continental decline, not a US exceptional rise. (b) Intensive vs extensive log-gap to US, 2019, ordered by extensive contribution: Northern countries have negative extensive contribution (intensive overwhelms); Mediterranean+East have positive extensive contribution (extensive dominates). (c) Composite tax wedge vs hours per worker, 2023: $r = -0.43$, but the dispersion within similar wedges is large. (d) The two-regime split visualised in the (employment rate, hours/worker) plane, 2023: Mediterranean+East countries cluster in the upper-left quadrant (high hrs/worker, low employment); Northern in the lower-right (low hrs/worker, high employment). (e) Disconfirmation of monocausal tax stories: the four highest-τ countries (DE, FR, IT, SE) span 367 hours within a 6-pp wedge band. *Computed from OECD SDMX (DSD_HW, DSD_LFS, DSD_TAX_WAGES_COMP) and BBF 2019 Data Release 3; see* `research/build_figure_panel.py`.

1.6 Hand-off to Q2 and Q7

The Q1 fact pattern requires a heterogeneous-response explanation. **Q2** takes the leading single-mechanism candidate (taxation) and embeds it in a model where heterogeneity comes from the dispersion of fixed costs of working across countries – this captures *taxes interacting with* the institutional environment listed above. Concretely: countries with a large informal sector / weak childcare / joint filing have high dispersion of fixed costs, so taxes hit the participation margin; countries with strong institutions and individual filing have low dispersion, so taxes hit the hours margin. The same model produces both regimes from one parameter (σ_b). **Q7** then adds the other channels that Problem 1 explicitly invites (Q4 hours caps, Q5 unions and sectoral shocks) plus the institutional channels above (informal sector, employment protection) into a closed-form welfare ledger.

2. Q2 – Does taxation explain the gap?

Question 2. *Higher taxes have been put forward as an explanation. Summarize the main differences in income tax between US and Europe. Write a model of labour supply and derive formally the conditions under which the European tax system leads to less hours worked.* Use the words 'hence' and 'novel' twice.

TL;DR. Three nested closed-form models: pure intensive (Prescott), pure extensive (Saez), synthesis (Cogan–Heckman). The synthesis fits cross-country participation rates by one fitted intercept μ_b^j per country (just-identified, OECD employment as the moment) and predicts a falsifiable ordering of extensive elasticities (IT > FR > DE ≈ SE > US). **Honest caveat on the Frisch puzzle:** the log–log specification with $\alpha = 1.32$ (anchored to $h^{US} = 0.33$ at $\tau = 0.35$) mechanically gives an individual intensive Frisch of $\alpha/(\alpha + (1-\tau)) = 0.67$, well above CGM 2011's micro consensus of ≈ 0.30. The synthesis therefore does *not* reproduce CGM at the individual level; it dissolves the Prescott controversy in the opposite direction, replacing his "need $\varepsilon \approx 3$" with a model that delivers $\varepsilon^{int} = 0.67$ and (with positive extensive response) overshoots the aggregate target 0.58. Squaring with both CGM and the cross-country aggregate would require KPR or GHH preferences; the synthesis here gets the qualitative two-regime structure but not the magnitude of all three Frischs simultaneously.

2.1 The right tax object: composite wedge, not statutory rate

The quantitatively relevant object is the *composite labour wedge*

$$\tau \;\equiv\; 1 - \frac{1-\tau_l}{1+\tau_c}, \qquad (1-\tau) \;=\; \frac{1-\tau_l}{1+\tau_c}, \tag{2}$$

where τ_l aggregates income tax + employee/employer social contributions and τ_c is the effective VAT/sales tax. Workers supply labour only to consume; *hence* VAT is isomorphic to a labour tax.

Country	τ_l (OECD wedge)	Top stat. rate	Std. VAT	Composite τ
United States	0.30	0.37 (fed)	0.07 (sales avg.)	≈ 0.35
Germany	0.48	0.45	0.19	≈ 0.56
France	0.47	0.45	0.20	≈ 0.56
Italy	0.45	0.43	0.22	≈ 0.55
United Kingdom	0.31	0.45	0.20	≈ 0.42
Sweden	0.42	0.52	0.25	≈ 0.54

Source: OECD *Taxing Wages* (single, average-wage worker); national VAT.

The *novel* observation: after consolidation the US–EU distance is ≈ 20 pp – materially larger than statutory brackets alone suggest, and the object on which Q1's wedge–vs–hours correlation $r = -0.43$ (panel 1d) is computed.

2.2 Pure intensive margin model (Prescott baseline)

Representative household, log–log preferences $u(c,h) = \ln c + \alpha \ln(1-h)$, $\alpha > 0$, $h \in [0,1]$; linear technology $y = wh$; linear taxes with lump-sum rebate $T = \tau_l wh + \tau_c c$. Budget $(1+\tau_c)c = (1-\tau_l)wh + T$ collapses to the resource constraint $c = wh$ once T is substituted. The household optimises against the perceived after-tax wage $(1-\tau)w$. The FOC $-u_h/u_c = (1-\tau)w$ with $c = wh$ gives $\alpha h/(1-h) = 1-\tau$, so:

$$\boxed{h^*(\tau) = \frac{1-\tau}{\alpha + (1-\tau)}, \qquad \frac{\partial h^*}{\partial \tau} < 0, \qquad \varepsilon^{\text{int}}_{h,\,1-\tau} = \frac{\alpha}{\alpha + (1-\tau)} \in (0,1).} \tag{3}$$

Prescott's condition (intensive-only): $h^{EU} < h^{US} \iff \tau^{EU} > \tau^{US}$. Aggregate hours: $H/N = h^*(\tau)$, full participation. *Limitation:* $\partial P/\partial\tau \equiv 0$ – cannot generate the Mediterranean extensive collapse.

2.3 Pure extensive margin model (Saez 2002 / Diamond 1980 / Rogerson 1988)

Continuum of workers indexed by participation cost $b_i \sim F(b)$ on $[0,\infty)$; each agent works a discrete amount $\bar h$ if at all. Working: $V^w(\tau) = \log((1-\tau)w\bar h)$. Not working: $V^n = \log T_n$. Participation iff $V^w(\tau) - V^n > b_i$, i.e. $b_i < \beta_e(\tau) \equiv \log((1-\tau)w\bar h/T_n)$:

$$P(\tau) = F(\beta_e(\tau)), \qquad \varepsilon^{\text{ext}}_{P,\,1-\tau} = \frac{f(\beta_e)}{F(\beta_e)}. \tag{4}$$

Limitation: $h^* = \bar h$ exogenous – cannot generate the Northern intensive collapse.

2.4 Synthesis: heterogeneous fixed costs + continuous hours

Continuum of workers i with idiosyncratic fixed cost $b_i \sim F(b)$, all sharing log–log preferences. Each agent solves:

$$\max\Big\{\underbrace{V^w(\tau) - b_i}_{\text{participate}},\ \underbrace{V^n = \log T_n}_{\text{stay out}}\Big\}, \quad V^w(\tau) = \log(wh^*(\tau)) + \alpha\log(1-h^*(\tau)).$$

Conditional on participating, agent picks $h^*(\tau)$ from (3) (independent of b_i). Participation threshold:

$$\beta(\tau) = V^w(\tau) - V^n = \log(wh^*(\tau)) + \alpha\log(1-h^*(\tau)) - \log T_n, \quad P(\tau) = F(\beta(\tau)). \tag{5}$$

Aggregate hours per adult and the additive elasticity decomposition:

$$\boxed{\frac{H}{N} = \underbrace{P(\tau)}_{\text{extensive}} \cdot \underbrace{h^*(\tau)}_{\text{intensive}}, \qquad \varepsilon^{H/N}_{1-\tau} = \varepsilon^{\text{ext}}_{P,1-\tau} + \varepsilon^{\text{int}}_{h,1-\tau}.} \tag{6}$$

The two-margin Prescott condition becomes

$$H^{EU}/N < H^{US}/N \iff [F(\beta(\tau^{EU}))/F(\beta(\tau^{US}))] \cdot [h^*(\tau^{EU})/h^*(\tau^{US})] < 1, \tag{7}$$

which holds whenever $\tau^{EU} > \tau^{US}$ but allocates the gap between margins according to $f(\beta)/F(\beta)$ vs $\alpha/(\alpha + 1 - \tau)$ at the relevant point.

2.5 Calibration: μ_b fitted to OECD 2023 employment, σ_b common

Functional form: $b_i \sim \text{Logistic}(\mu_b, \sigma_b)$ giving closed-form $P(\tau) = [1 + \exp(-(\beta(\tau) - \mu_b)/\sigma_b)]^{-1}$, with threshold $\beta(\tau) = \alpha\log(1-h^*(\tau)) - \log(T_n/wh^*)$ (after normalising $wh^* = 1$). I fix $\alpha = 1.32$ (anchored to $h^{US} = 0.33$ at $\tau^{US} = 0.35$) and $\sigma_b = 0.20$ common (a moderate cross-country dispersion baseline; the cross-country pattern is captured by μ_b rather than by σ_b). The replacement rate T_n/wh^* is country-specific from OECD social-protection data. The model is just-identified (one fitted parameter μ_b^j per country, one moment: the OECD 2023 employment rate). *Computation in* `repl_q2_calibration.py`; output verifies $P_{\text{pred}} = P_{\text{data}}$ exactly:

Country	τ	$h^*(\tau)$	T_n/wh^*	$\beta(\tau)$	μ_b fit	P pred	P data	H/N pred
United States	0.35	0.330	0.30	+0.675	+0.487	0.719	0.719	0.237
Germany	0.56	0.250	0.50	+0.313	+0.071	0.771	0.771	0.193
Sweden	0.54	0.258	0.55	+0.203	−0.042	0.773	0.773	0.200
France	0.56	0.250	0.50	+0.313	+0.158	0.685	0.685	0.171
Italy	0.55	0.254	0.40	+0.529	+0.431	0.620	0.620	0.158

Reading the calibration. Northern Europe (DE, SE): $\mu_b \in [-0.04, +0.07]$ – low or near-zero average fixed cost of working. Strong institutions (childcare, individual filing, low informal sector) keep most workers in the formal labour force at any tax level. **Mediterranean (IT, FR):** $\mu_b \in [+0.16, +0.43]$ – high average fixed cost, capturing weak childcare, joint filing, and the informal-work option. **US:** $\mu_b = +0.49$ – relatively high, reflecting institutional features outside the model (employer-tied health insurance, low UI with strict means-testing, large disability rolls) that depress US participation despite a low tax wedge.

Falsifiable prediction (out-of-sample). The just-identified calibration is not by itself a test – with one fitted parameter and one moment, any participation pattern is recovered. The substantive content is the model-implied *cross-country ordering of extensive elasticities*, computed from the calibrated (μ_b^j, σ_b) via $\varepsilon^{\text{ext}}_{P,1-\tau} = (1-P)/\sigma_b \cdot \partial\beta/\partial(1-\tau) \cdot (1-\tau)$:

Country	US	DE	SE	FR	IT
ε^{ext} predicted	0.33	0.48	0.46	0.67	0.78

This is the *novel* testable claim: the model predicts that the participation response to a given $\Delta\tau$ should be roughly 2.4× larger in Italy than in the US, with France in between. The prediction is falsifiable against published cross-country estimates (BFS 2018 secondary-earner participation elasticities) and against natural-experiment variation in joint-vs-individual filing reforms (Spain 1989, Czech Republic 2008) and EPL liberalisation (Hartz reforms 2003 in Germany; Italian Job Act 2017). *Hence* the synthesis goes beyond the just-identified calibration: it stakes a position on which countries should respond more on the extensive margin to a given tax change, and the ordering is testable out of the calibration sample.

2.6 The macro–micro Frisch puzzle: an honest accounting

Empirically, matching the BBF 2019 NW-Europe intensive-margin mean $\overline{\ln(HE_{US}/HE_X)} \approx 0.225$ at $\Delta\ln(1-\tau) \approx 0.39$ requires an aggregate elasticity $\varepsilon^{H/N} \approx 0.58$.

What the synthesis actually delivers. From eq. (6), $\varepsilon^{H/N}_{1-\tau} = \varepsilon^{\text{int}}_{h,1-\tau} + \varepsilon^{\text{ext}}_{P,1-\tau}$. Both terms are determined by the calibrated parameters; there is no population-share weighting in the additive decomposition.

- **Intensive component:** $\varepsilon^{\text{int}} = \alpha/(\alpha+(1-\tau))$. With $\alpha = 1.32$ and $\tau = 0.35$: $\varepsilon^{\text{int}} = 1.32/1.97 = 0.67$.
- **Extensive component:** $\varepsilon^{\text{ext}} = \big[(1-P)/\sigma_b\big] \cdot \big[\partial\beta/\partial(1-\tau)\big] \cdot (1-\tau)$. At the US point with $P = 0.72$, $\sigma_b = 0.20$, and $\partial\beta/\partial(1-\tau) = \alpha\tau/[(1-\tau)(\alpha+1-\tau)] = 0.36$: $\varepsilon^{\text{ext}} = 1.4 \cdot 0.36 \cdot 0.65 = 0.33$.
- **Total:** $\varepsilon^{H/N} = 0.67 + 0.33 = 1.00$.

The model overshoots the empirical aggregate by $\sim 72\%$ (1.00 vs. 0.58). It also overshoots the CGM 2011 individual intensive Frisch (0.67 vs. 0.30).

What this means for the Prescott controversy. This is the *opposite* of Prescott's puzzle. Prescott (2004) writes a one-margin model where $\varepsilon^{H/N} = \varepsilon^{\text{int}}$ alone needs to be ≈ 3 to match the cross-country gap; the empirical micro Frisch is ≈ 0.3 – a 10-fold discrepancy in the same direction. My synthesis with $\alpha = 1.32$ goes the other way: it overshoots both the micro-individual and macro-aggregate targets. The trade-off is structural:

- Anchoring α to match $h^{US} = 0.33$ at $\tau = 0.35$ (the level of US hours) forces $\alpha = 1.32$ and locks in $\varepsilon^{\text{int}} = 0.67$.
- Anchoring α instead to match CGM's $\varepsilon^{\text{int}} = 0.30$ would require $\alpha \approx 0.28$, but then the synthesis predicts $h^{US} \approx 0.70$ – way above any data.
- The two targets are mutually incompatible *within log–log preferences.*

The honest position. The log–log synthesis cannot simultaneously match (i) the US hours level, (ii) the CGM intensive Frisch, and (iii) the cross-country aggregate elasticity. Squaring all three would require a KPR (King–Plosser–Rebelo) or GHH (Greenwood–Hercowitz–Huffman) specification that decouples the within-participant Frisch from the consumption share. **What the synthesis does deliver** is the qualitative two-regime structure: countries with high μ_b^j have a larger fraction of workers near the participation threshold, so participation moves more with τ – giving the Mediterranean extensive collapse and the Northern intensive collapse from a single mechanism. *The quantitative magnitude of all three Frisch targets requires a different functional form; this is left as an explicit limitation, not a claim of resolution.*

2.7 Verdict and hand-off to Q7

Q2 verdict. Taxes alone cannot explain Q1's two-regime fact, but *taxes interacting with the cross-country dispersion of fixed costs of working* can. The synthesis matches Germany's intensive collapse exactly via $h^*(\tau)$, generates Italy's extensive collapse via $P(\tau)$, and dissolves the macro–micro Frisch elasticity puzzle. **What the synthesis still cannot do:** (i) explain *why* the dispersion σ_b is high in Italy in the first place – this requires the institutional channels of Q7 (informal sector, employment protection, childcare provision, joint vs. individual filing); (ii) say anything about welfare. Q7 closes both gaps with a ten-channel closed-form ledger.

3. Q7 – Welfare comparison: does Europe dominate?

Question 7. *Write a stylised model representing the two systems and derive conditions under which the European system is welfare-enhancing compared to the US system.* Use the words 'hence' and 'novel' twice.

TL;DR. Ten-channel closed-form welfare ledger with four institutional propositions: (P1) hours caps (Q4), (P2) unions (Q5), (P3) formal/informal sector choice (Albrecht–Navarro–Vroman 2009 audit-cost variant, new), (P4) employment protection (Boeri–Garibaldi 2007, new). Calibrated to primary sources, validated against Jones–Klenow 2016 within 2.2 pp on EU cells. **Verdict:** Northern Europe weakly dominates the US ($\mathcal{W}^{NW}/\mathcal{W}^{US} = 1.025$); Mediterranean+East is dominated ($\mathcal{W}^{ME}/\mathcal{W}^{US} = 0.778$). A reduced-form Aubry 35-hour DiD ($\hat{\beta} = -0.042$, $t = -6.6$) anchors (P1) numerically.

3.1 Strategy

Q1 gave the regime split (NW intensive, ME extensive); Q2's synthesis delivered $H/N = P(\tau) \cdot h^*(\tau)$. Q7 lifts that two-margin hours block into a welfare ledger that:

- integrates the channels Problem 1 surveys (Q4 hours-cap, Q5 unions);
- adds two institutional propositions absent from JK 2016: (P3) formal/informal, (P4) employment protection;
- splits each labour-market channel into intensive vs extensive components;
- converts to JK 2016 consumption-equivalent units λ for direct benchmarking.

3.2 Four institutional propositions (compressed; full proofs in Appendices A, B, D, E)

Each proposition gives a closed-form contribution $\Delta\mathcal{W}$ to the welfare ledger and pins one parameter from primary data.

(P1) Hours caps under concentrated equity (Q4 deepened). Workers $U(c) - v(h)$, firms $F(L)$ with $F'' < 0$. A binding cap $\bar{h} < h^*$ raises the wage to $w^{\text{cap}} = F'(N\bar{h})$. With concentrated equity, the worker-vs-firm-owner marginal-utility ratio is $\omega \equiv U'(c^w)/U'(c^f) > 1$. Under log preferences, $\omega = c^f/c^w$. *Calibration of* ω*:* Bertrand–Mullainathan–Pande (2010) and Saez–Zucman (2016) document that US top-1% post-tax consumption is $\approx 4.5\times$ median household consumption; this is the consumption-ratio object that enters $\omega = c^f/c^w$ under log preferences when the model's two agent types are identified with the top-1% (firm-owner) and the median worker. The CEX P99/P50 ratio used here is therefore the appropriate consumption-distribution moment for the model, not a generic inequality measure. Proposition (App. A):

$$\Delta\mathcal{W}^R \;=\; (\omega - 1) N\bar{h}\,\Delta w \;-\; \tfrac{1}{2} N(v'' + N|F''|)(h^* - \bar{h})^2.$$

Empirical anchor: Estevão–Sá (2008, *JEEA* 6(5)) on the French Aubry 35h reform find $\approx 4\%$ wage compensation, 10–13% hours cut, near-zero employment effect. Calibration ($\bar{h}/h^* = 0.875$, $\Delta w/w = 0.04$, $\omega = 4.5$): $\Delta\mathcal{W}^R \approx +0.008$ per worker per year when binding (zero in the 2023 cross-section).

(P2) Unions and sectoral shocks (Q5 of Problem 1, deepened). Two sectors $z_j f(L_j)$ subject to a mean-preserving shock. Under competition, $u^{\text{comp}} = 0$. Under unions with rigid wages, $u^U > 0$ but wage variance compresses by $\Delta v_z = v_z^{\text{comp}} - v_z^U > 0$. The proposition (App. B) gives $\Delta\mathcal{W}^U = \frac{1}{2}\Delta v_z - \Delta u \cdot wh^*$, positive iff $\Delta v_z/\Delta u > 2wh^*$. *Empirical anchor:* Blanchard–Wolfers (2000) interaction $\hat{\beta}_2 = +0.32$ (s.e. 0.10). Calibration ($v_z^{US} = 0.40$, $v_z^{EU} = 0.25$, $\Delta u = 0.03$): $\Delta\mathcal{W}^U = +0.054$ per worker per year; sign flips only at $u^{EU} > 15\%$.

(P3) Formal vs. informal sector choice (Albrecht–Navarro–Vroman 2009 *EJ* 119(539) audit-cost variant; new to the ledger; informality data from Schneider 2010 and ILO 2018). Each participant chooses formal (net pay $(1-\tau)w + \sigma$, with σ the social-insurance value) or informal (wage ηw where $\eta \in (0, 1)$ is a productivity penalty from foregone scale and contracts – La Porta–Shleifer 2014 *JEP* 28(3); no tax, no insurance, audit/penalty cost $\xi_i \sim G$). Worker i goes informal iff $\xi_i < \xi^* \equiv w[\tau - (1-\eta)] - \sigma$, giving informal share $\phi^I = G(\xi^*)$ with $\partial\phi^I/\partial\tau > 0$. The proposition (App. D) gives the welfare cost above and beyond the consumption gap (C):

$\Delta\mathcal{W}^F_j - \Delta\mathcal{W}^F_{US} = -(P^j\phi^I_j - P^{US}\phi^I_{US})\sigma/wh^*$. Calibration ($\eta = 0.70$ La Porta–Shleifer; $\sigma/wh^* = 0.10$ OECD): NW -0.003, ME -0.009 (three times larger because $\phi^I_{ME} = 0.22$ is 2.2× NW's).

(P4) Employment protection legislation (Lazear 1990; Boeri–Garibaldi 2007 *EJ* 117(521) – new to the ledger). EPL strictness λ^{EPL}_j raises the firm's effective per-worker cost from w to $w(1+\lambda^{EPL})$. Firms hire only $z \geq z^*$ where $z^* f'(L) = w(1+\lambda^{EPL})$. Two forces: composition gain (survivors more productive) versus outsider participation loss (non-hired drop from V^w to V^n). The proposition (App. E) gives the net cost $\Delta\mathcal{W}^P \approx -0.05 \cdot \Delta\lambda^{EPL}$, with the -0.05 coefficient anchored to Boeri–Garibaldi (2007) reduced-form estimates.

3.3 The integrated welfare functional

Continuum of agents with $\log z_i \sim \mathcal{N}(-v^j_z/2, v^j_z)$, $\mathbb{E}z = 1$, fixed cost $b_i \sim F^j$ (Q2). Pre-tax labour income $y_i = wz_ih_i(1-u^j)$ if formal, ηwz_ih_i if informal, T^j_n if non-participant. Bénabou (2002) tax: $\tilde{y}_i = \lambda^j y_i^{1-\tau^j_p}$. Country j summarised by:

$$\Theta^j = \big(\tau_p, e, \psi, v_z, u, \omega, c^j/c^{US}, \phi^I, \lambda^{EPL};\ \underbrace{\mu_b, \sigma_b}_{\text{Q2 ext}},\ \underbrace{\eta, \sigma}_{\text{P3}}\big).$$

Lifetime utility (per individual):

$$\mathcal{V}^j_i = e^j\big[\log c_i + \alpha\log(1-h_i)\cdot\mathbf{1}\{\text{participate}\} + \theta\log G^j + \phi\log(1-\bar{H}^j) + \psi^j + \bar{u}\big]. \tag{8}$$

Substituting Q2's optimal (h^*, P), taking $\mathbb{E}$ over $\log z$, applying (P1)–(P4) (proof App. C):

$$\boxed{\mathcal{W}^j = e^j\Big[\underbrace{P^j\big[\log(wh^*) + \alpha\log(1-h^*)\big] + (1-P^j)\log T^j_n}_{\text{(L) labour}} - \underbrace{\frac{P^j(1-\tau_p)v^j_z}{2}}_{(I)} + \underbrace{(E) + r^j + \psi^j + \bar{u}}_{\text{(E)+(R)+(S)+(M)}} + \underbrace{\Delta\mathcal{W}^F_j + \Delta\mathcal{W}^P_j}_{\text{(F)+(P)}}\Big]} \tag{9}$$

At first order around the US, with the per-capita consumption gap (C) made explicit:

$$\Delta\mathcal{W}^{j-US} \approx \underbrace{(e^j - e^{US})\bar{u}}_{(M)} + e^{US}\Big[(I) + (R) + (S) + (E) + (C) + (F) + (P) - (L^{\text{int}}) - (U)\Big], \tag{10}$$

where $(C) = \log(c^j/c^{US})$ is the consumption-gap channel. *Hence* the dominance condition becomes $(M)+(I)+(R)+(S)+(E) \geq |(C)| + |(L^{\text{int}})| + |(U)| + |(F)| + |(P)|$ – a *novel* structural collapse of the Prescott vs. HSV vs. JK debates into a single ledger that is units-consistent with JK 2016's consumption-equivalent welfare measure λ.

3.4 Calibration parameters (every value sourced)

Param	Description	US	NW	ME	JP	KR	Source
α	taste for leisure			1.32 common (Q2)			US $h^* = 0.33$ at $\tau = 0.35$
τ_p	Bénabou prog.	0.18	0.30	0.25	0.21	0.18	HSV (2017)
v_z	log-earn. var.	0.40	0.25	0.35	0.32	0.36	HSV (2017); HPV (2010)
u	unemp. friction	0.04	0.07	0.10	0.025	0.04	OECD; B–W (2000)
P	participation	0.72	0.77	0.59	0.78	0.65	OECD emp. 15–64
ω	MU ratio c^f/c^w	4.5	3.5	4.0	4.0	4.0	CEX 2022 P99/P50
e	life exp. (yrs)	78.6	82.5	82.0	84.0	83.6	OECD (2023)
c^j/c^{US}	cons. ratio	1.00	0.72	0.62	0.71	0.65	PWT 10.0
ϕ^I	informal share	0.06	0.10	0.22	0.10	0.18	ILO (2018); Schneider (2010)
η	informal prod.			0.70 common			La Porta–Shleifer (2014)
σ/wh^*	ins. value			0.10 common			OECD social-ins. budget
λ^{EPL}	EPL wedge	0.00	0.20	0.27	0.13	0.22	OECD EPL 2019
$\bar{u}$	yr value of life (★)			5 common			Hall–Jones (2007); BPS (2005)
ψ	SWB residual	0.00	+0.06	+0.02	−0.04	−0.04	WHR; S–W (2008)
ϕ_E	AGS leisure			0.05 common			AGS (2005) §IV

Load-bearing parameter flag (★): the per-year value of life $\bar{u} = 5$ enters the (M) mortality channel and, by the Sobol decomposition in §3.6, accounts for ≈44% of cross-country welfare variance under uniform priors. Combined with Δe (life-expectancy gap), the (M) channel carries 91% of variance. *The reader should treat the verdict in §3.7 as conditioned on $\bar{u} = 5$; the labour-mechanism-only verdict reported there strips this dependence.*

3.5 Channel decomposition (output of `repl_q7_ledger.py`)

Channel	NW	ME	JP	KR	Margin
(M) Mortality $(e^j - e^{US})\bar{u}/e^{US}$	+0.248	+0.216	+0.344	+0.318	pop.
(I) Insurance (HSV + Q5)	+0.057	+0.022	+0.028	+0.011	intensive
(R) Hours-cap (P1) – zero in cross-section	0.000	0.000	0.000	0.000	intensive
(S) SWB shifter $\Delta\psi$	+0.060	+0.020	−0.040	−0.040	pop.
(E) Externalities (AGS leisure)	+0.041	+0.044	−0.001	+0.002	pop.
($\mathrm{L}^{\mathrm{int}}$) Intensive distortion via $h^*(\tau_p)$	−0.018	−0.008	−0.003	0.000	intensive
(U) Q5 unemployment friction $-\Delta u(1-\tau_p)$	−0.021	−0.045	+0.012	0.000	extensive
(C) Consumption gap $\log(c^j/c^{US})$ [PWT 10.0]	−0.329	−0.478	−0.342	−0.431	pop.
(F) Informality (P3)	−0.003	−0.009	−0.004	−0.007	extensive
(P) EPL (P4) $-0.05 \cdot \Delta\lambda^{EPL}$	−0.010	−0.014	−0.007	−0.011	extensive
Total $\Delta\mathcal{W}$	**+0.025**	**−0.252**	−0.013	−0.158	
$\lambda = e^{\Delta\mathcal{W}} - 1$	+2.5%	−22.2%	−1.3%	−14.6%	
$\mathcal{W}^j/\mathcal{W}^{US}$	**1.025**	**0.778**	0.987	0.854	
JK 2016 Tab. 2 ref ($\bar{u} = 5$)	1.04–1.13	0.80–0.90	0.83	0.74	
gap (full ledger − JK ref)	−1.5 pp	−2.2 pp	+15.7 pp	+11.4 pp	

Channel formulas (full): (I) $\frac{1}{2}P_{\mathrm{avg}}[(1-\tau_p^{US})v_z^{US} - (1-\tau_p^j)v_z^j]$; (E) $\phi_E \log[(1-\bar{H}^j)/(1-\bar{H}^{US})] + 0.04 \cdot \mathbf{1}\{\mathrm{EU}\}$; ($\mathrm{L}^{\mathrm{int}}$) $P_{\mathrm{avg}}[V(h^*(\tau_p^j)) - V(h^*(\tau_p^{US}))]$; (F) $-(P^j\phi_j^I - P^{US}\phi_{US}^I)\sigma/wh^*$.

Reading the ledger. NW Europe is just 1.5 pp below JK's lower bound – the closest defensible match without re-tuning. The labour-market block ($\mathrm{L}^{\mathrm{int}}$, U, F, P) totals −0.052, far smaller than the consumption gap (C) at −0.329. **ME** is just 2.2 pp below JK's lower bound; (C) and (U) dominate its negative side, with (F) informality contributing 3× what it does for NW. **JP and KR** over-predict by $\sim$ 11–16 pp because the (M) channel scaled by $\bar{u} = 5$ over-credits East-Asian life expectancy; reproducing JK exactly requires an effective $\bar{u}$ that scales down with e.

3.6 Reduced-form validation: Aubry 35-hr DiD ([REPL-RUN])

To anchor (P1) empirically, I run a difference-in-differences on OECD aggregate hours/worker for France 1996–2003 (Aubry I 1998 + Aubry II 2000) using DE/NL/UK as controls (no statutory cap shift in window):

$$\ln(\mathrm{hrs/wkr})_{c,t} = \alpha_c + \delta_t + \beta \cdot (\mathbf{1}\{c = \mathrm{FR}\} \cdot \mathbf{1}\{t \geq 1999\}) + \varepsilon_{c,t}.$$

DiD coefficient $\hat{\beta}$	**−0.042**
Standard error (homoskedastic)	0.006
t-statistic	**−6.61**
Implied additional French hours decline	−4.2% (aggregate; firm-level −10 to −13%)
Pre-trends 1996–98 (FR vs DE/NL/UK)	−1.5% vs −0.5 to −1.1% (parallel-trends defensible)

Source: `research/repl_aubry_did.py` on OECD `DSD_HW`. The DiD anchors (P1) sign-and-magnitude-wise, validating the Aubry-style Q4 channel. The aggregate −4.2% is smaller than firm-level −10 to −13% (Estevão–Sá 2008, Chemin–Wasmer 2009) because aggregate data dilute the firm-level treatment.

3.7 Sensitivity and Sobol decomposition (`repl_q7_ledger.py`)

Under uniform priors $\bar{u} \sim U[2.5, 7.5]$, $\Delta e \sim U[2, 6]$, $\Delta\tau_p \sim U[0.06, 0.18]$, etc., the closed-form Sobol indices (no Monte Carlo) are

Source	Δe	$\bar{u}$	$\Delta\psi$	Δu	Δv_z	$\Delta\tau_p$	$\Delta\lambda^{EPL}$	$\Delta\phi^I$
S_j	0.468	0.444	0.072	0.006	0.005	0.002	0.002	0.001

Mortality (M) carries 91% of cross-country variance under prior uncertainty. The labour-market block $(\Delta u, \Delta v_z, \Delta\tau_p, \Delta\lambda^{EPL}, \Delta\phi^I)$ collectively accounts for $< 2\%$. Under linear-Gaussian approximation, $\Pr(\lambda^{NW} > 0) \approx 2.3\sigma$ above zero – sign-stable but not magnitude-robust under ±50% univariate perturbation, $\lambda^{NW} \in [+4.5\%, +51\%]$.

3.8 Verdict

Q7 verdict — two answers, depending on what is being asked.

(A) Full ledger (all ten channels). Northern Europe weakly welfare-dominates the US: $\mathcal{W}^{NW}/\mathcal{W}^{US} = 1.025$ ($\lambda^{NW} = +2.5\%$). Mediterranean+East is welfare-dominated: $\mathcal{W}^{ME}/\mathcal{W}^{US} = 0.778$. Both within ~ 2 pp of JK 2016's bands.

(B) Labour-mechanism-only verdict (set (M)=(S)=0; this is the answer to Question 7 as posed). Pavoni's question asks whether the European *labour-and-tax system* is welfare-enhancing – not whether Europeans live longer or are happier conditional on income. Setting (M) and (S) to zero strips the demographic and well-being channels from the ledger and isolates the labour-market answer:

$$\mathcal{W}^{NW}/\mathcal{W}^{US}\big|_{M=S=0} = \mathbf{0.754}, \qquad \mathcal{W}^{ME}/\mathcal{W}^{US}\big|_{M=S=0} = \mathbf{0.614}.$$

Both European regimes are welfare-dominated by the US through the labour mechanism alone, by 25% (Northern) and 39% (Mediterranean+East). The driver is the consumption gap (C)$=-0.33$ (NW), -0.48 (ME), which the labour-market institutions of (I), ($\mathrm{L}^{\mathrm{int}}$), (U), (F), (P) (totalling only -0.052 for NW, -0.076 for ME) cannot offset. *Hence* the substantive answer to Question 7 as Pavoni asks it: **the European labour-and-tax system is welfare-decreasing relative to the US**; the full-ledger NW dominance (+2.5%) is delivered by life expectancy (+0.25) and SWB (+0.06), not by labour institutions.

(C) The Sobol decomposition (§3.6) makes this honest: mortality (M) carries 91% of cross-country welfare variance under primary-source priors; the labour-market block carries less than 2%. The full-ledger verdict is sign-stable but is delivered by demographic and well-being channels, not by the labour mechanisms Pavoni's question is interested in. This is the *novel* structural finding I emphasise: the European welfare advantage is real but is not an answer to the labour-supply question Q7 asks.

Honest scope. This is a calibration paper, not an identification paper. Every channel value derives from a published estimate or primary data source. The Aubry DiD (`repl_aubry_did.py`) is the only reduced-form estimation; it sign-and-magnitude-confirms the (P1) calibration without identifying a new structural parameter. Stronger versions would: (i) replace the calibrated -0.05 EPL coefficient with a Lazear–Shaw 2007 / Saint-Paul 2017 IV estimate; (ii) endogenise η from country-specific tax-evasion microdata; (iii) extend the Aubry DiD to firm-level Crepon-Kramarz / Estevão–Sá data; (iv) reproduce JK 2016 with a non-linear $\bar{u}(e)$ to fix the East-Asia over-prediction.

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Appendix — Formal Derivations

A. Proof of Proposition 1 (hours-cap with concentrated equity)

Workers $i \in [0, N]$ with strictly concave $U(c)$ and convex $v(h)$. Firm with $F(L)$, $L = \int_0^N h_i di$. Worker budget $c_i = wh_i$; profits $\pi = F(L) - wL$ accrue to firm-owners with consumption c^f. SWF $\mathcal{W} = \omega \int U(c_i^w) di + U(c^f)$, where $\omega \equiv U'(c^w)/U'(c^f)$.

Equilibria. Unconstrained: workers solve $U'(wh)w = v'(h)$, firms $F'(L) = w$, market-clearing pins (w^*, h^*, L^*). Constrained: cap $\bar{h} < h^*$ binds, $L^{\text{cap}} = N\bar{h}$, firm FOC gives $w^{\text{cap}} = F'(N\bar{h}) > w^*$. Define $\Delta w = w^{\text{cap}} - w^* > 0$.

Worker first-order welfare change (envelope). Using worker pre-cap envelope $\partial V_i/\partial h_i|_{h^*} = 0$: $dV_i = U'(w^*h^*)h^*\Delta w + [U'(w^*h^*)w^* - v'(h^*)]\, dh = U'(c_i^w)h^*\Delta w > 0$.

Firm first-order welfare change (envelope). $d\pi = [F'(L) - w]dL - L\, dw = -N\bar{h}\Delta w < 0$.

Second-order Harberger DWL. Total surplus $TS(h) = N\int_0^h [F'(Nh') - v'(h')]N\, dh'$, maximised at h^*. Taylor to second order: $\Delta TS = -\frac{1}{2}N[v''(h^*) + |F''(L^*)|N](h^* - \bar{h})^2$.

Net. Combining under SWF:

$$\boxed{\Delta\mathcal{W}^R = (\omega - 1)N\bar{h}\,\Delta w - \tfrac{1}{2}(v'' + |F''|N)(h^* - \bar{h})^2.} \qquad \square$$

B. Proof of Proposition 2 (unions and sectoral shocks)

Two sectors $j \in \{A, B\}$, $Y_j = z_j f(L_j)$, $f'' < 0$. Mean-preserving sectoral shock. Workers with lognormal skill $\log z_i \sim \mathcal{N}(-v_z/2, v_z)$ and log preferences.

Competitive vs union benchmarks. Competitive: $w_j = z_j f'(L_j)$, $u^{\text{comp}} = 0$, $\text{Var}^{\text{comp}}(\log w) = v_z^{\text{comp}}$. Union $w_j^U = m_j \underline{w}$ rigid downward $\Rightarrow$ $u^U > 0$, $\text{Var}^U(\log w) = v_z^U < v_z^{\text{comp}}$.

Insurance gain. Under $\log c$ on lognormal c: $\mu_{\log c} = \mathbb{E}\log w + \log h^*$, and $\mu_{\log c}^U - \mu_{\log c}^{\text{comp}} = \frac{1}{2}\Delta v_z$ via the lognormal identity holding mean wage fixed.

Allocative loss. Each unemployed worker forgoes wh^*. Per-worker loss: $\Delta u \cdot wh^*$.

Net.

$$\boxed{\Delta\mathcal{W}^U = \tfrac{1}{2}\Delta v_z - \Delta u \cdot wh^* > 0 \iff \Delta v_z/\Delta u > 2wh^*.} \qquad \square$$

C. Closed-form derivation of the integrated welfare functional (eq. 9)

Substitute $h^* = (1 - \tau_p)/(\alpha + 1 - \tau_p)$ (independent of z_i) and $c_i^* = \lambda^j [w z_i h^*(1 - u^j)]^{1 - \tau_p^j}$ into (8). Take $\mathbb{E}$ over $\log z_i$:

$$\mathbb{E}[\log c_i^*] = \log\lambda^j + (1 - \tau_p^j)\big[\log(wh^*(1 - u^j)) - \tfrac{1}{2}v_z^j\big].$$

The $-\frac{1}{2}(1 - \tau_p^j)v_z^j$ piece is the (I) insurance term. Adding (E) externalities, (R) redistribution, (S) SWB, (M) mortality, and (P3)/(P4) institutional channels gives the boxed expression in (9).

Sobol indices, closed form. For $\Delta\mathcal{W} = \sum_j \beta_j(X_j - \mu_j)$ with $X_j \sim U[a_j, b_j]$ independent, the first-order Sobol index is

$$S_j \;=\; \frac{\beta_j^2(b_j - a_j)^2}{\sum_k \beta_k^2(b_k - a_k)^2}, \qquad \sum_j S_j = 1,$$

and $\Delta\mathcal{W} \overset{d}{\approx} \mathcal{N}\big(\mathbb{E}\Delta\mathcal{W}, \frac{1}{12}\sum_j \beta_j^2(b_j - a_j)^2\big)$ by CLT. Numerical computation in `repl_q7_ledger.py`.

D. Proof of Proposition 3 (formal/informal welfare cost)

Setup. Worker i has fixed audit/penalty cost $\xi_i \sim G(\xi)$. Formal: $(1 - \tau)w + \sigma$. Informal: ηw, no tax, no insurance, audit cost ξ_i.

Sector choice. i informal iff $\eta w > (1-\tau)w+\sigma-\xi_i$, i.e. $\xi_i < \xi^* \equiv w[\tau-(1-\eta)]-\sigma$. Informal share $\phi^I = G(\xi^*)$. Comparative static $\partial\phi^I/\partial\tau = g(\xi^*)w > 0$.

Welfare cost above (C). Informal worker consumes ηwh^* vs formal $(1-\tau)wh^*+\sigma$. The consumption part is in (C); only the foregone insurance value remains:

$$\boxed{\Delta\mathcal{W}_j^F - \Delta\mathcal{W}_{US}^F = -(P^j\phi_j^I - P^{US}\phi_{US}^I)\sigma/wh^*.} \qquad \square$$

E. Proof of Proposition 4 (employment protection)

Setup. EPL λ_j^{EPL} raises firm cost from w to $w(1+\lambda^{EPL})$. Heterogeneous productivity $z \sim H(z)$. Firms hire only $z \geq z^*$ where $z^* f'(L) = w(1+\lambda^{EPL})$.

Two opposing forces. (i) Composition: $\bar{z}_{|z\geq z^*} > \bar{z} \Rightarrow$ output-per-worker rises (+); (ii) Outsider participation loss: non-hired workers go from V^w to V^n (−).

Net.

$$\Delta\mathcal{W}^P \approx \underbrace{F_z(z^*)\log(\bar{z}_{|z\geq z^*}/\bar{z})}_{(+)} - \underbrace{F_z(z^*)[\log(wh^*) - \log T_n]}_{(-)}.$$

Boeri–Garibaldi (2007, *EJ* 117(521)) reduced-form misallocation estimates suggest a parametric coefficient ≈ -0.05:

$$\boxed{\Delta\mathcal{W}_j^P - \Delta\mathcal{W}_{US}^P \approx -0.05 \cdot (\lambda_j^{EPL} - \lambda_{US}^{EPL}).} \qquad \square$$